

COMSATS University Abbottabad

Semester Project Assignment – 02

Course: Game Development

CLO-2: Create different assets and scenes for a game scenario.

Bloom's Taxonomy Level: Creating (3,4)

Assignment Type: Semester Project – Phase 1

Total Marks: 50

Duration: 1 Week

Project Title

**Design and Development of a Multi-Environment Game Prototype
(BlackForest - Survival Horror)**

Submitted By:

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Problem Statement & Scenario

As a Game Environment Designer, the goal of this assignment is to develop the visual and interactive foundation of 'BlackForest'. Utilizing assets from Assignment-01, this document outlines the construction of a fully playable level containing interactive mechanics, environmental storytelling, and audio integration, ensuring a 5-10 minute gameplay experience.

Task 1: Gameplay System Design

1.1 Player Movement System

Explanation: The player movement system utilizes Unity's Rigidbody2D for physics-based movement. A virtual joystick or standard mobile touch input translates to a 2D movement vector.

```
using UnityEngine;

public class PlayerMovement : MonoBehaviour
{
    public float moveSpeed = 5f;
    private Rigidbody2D rb;
    private Vector2 movement;

    void Start() { rb = GetComponent<Rigidbody2D>(); }

    void Update()
    {
        movement.x = SimpleInput.GetAxis("Horizontal");
        movement.y = SimpleInput.GetAxis("Vertical");
    }

    void FixedUpdate()
    {
        rb.MovePosition(rb.position + movement * moveSpeed *
Time.fixedDeltaTime);
    }
}
```

[Insert Screenshot: Player GameObject in Unity Inspector showing Rigidbody2D and Movement Script]

1.2 Interaction & Objective System (Campfire & Wood)

Explanation: The core objective is keeping the fire alive. The player interacts with trees to get wood and deposits it into the fire using triggers (OnTriggerEnter2D).

```
using UnityEngine;

public class CampfireInteraction : MonoBehaviour
{
    public float fireFuel = 100f;
```

```

public float depletionRate = 2f;
public UnityEngine.Rendering.Universal.Light2D fireLight;

void Update()
{
    fireFuel -= depletionRate * Time.deltaTime;
    fireLight.pointLightOuterRadius = fireFuel * 0.05f;
    if(fireFuel <= 0) { GameManager.Instance.GameOver(); }
}

private void OnTriggerStay2D(Collider2D collision)
{
    if (collision.CompareTag("Player") && Input.GetButtonDown("Interact"))
    {
        if (PlayerInventory.woodCount > 0)
        {
            fireFuel += 20f;
            PlayerInventory.woodCount--;
        }
    }
}
}

```

[Insert Screenshot: Gameplay Interface showing Fire Fuel Bar and Wood Inventory]

1.3 Gameplay Flow Diagram

[Insert Gameplay Flow Diagram Image Here]

Task 2: Interactive Environment Construction

As per the assignment requirements, the scenes have been populated with interactive elements to transform the static environment.

2.1 Required Elements Inventory

Interactive Objects (Minimum 5):

1. Campfire (Fuel interaction)
2. Choppable Trees
3. Locked Cabin Door (Requires key)
4. Hidden Switch (Opens shortcuts)
5. Safezone Checkpoint (Saves progress)

Collectible Items (Minimum 5):

1. Wood Logs
2. Matchboxes (Instant fire revive)
3. Berries (Restore health/energy)
4. Lost Journal Pages (Lore)
5. Speed Boots (Temporary movement buff)

Obstacles/Traps (Minimum 3):

1. Shadow Enemies (Chasing AI)
2. Bear Traps hidden in bushes
3. Poisonous Spores (Area of effect damage)

Environmental Effects (Minimum 3):

1. Dynamic URP Fog Particle System
2. Global 2D Lighting (Night cycle)
3. Rain/Storm particle system

Trigger Events (Minimum 4):

1. Shadow Spawn Trigger (Based on survival time)
2. Low Health Vignette Trigger
3. Cabin Door Unlock Event
4. Game Over UI Event

[Insert Screenshot: Unity Scene View showing placed items, traps, and triggers]

Task 3: Audio and Visual Enhancement

Audio is crucial for the survival horror atmosphere of BlackForest. The AudioManager script handles background ambience and interaction sounds.

```
using UnityEngine;
```

```
public class AudioManager : MonoBehaviour
{
    public AudioSource bgmSource;
    public AudioSource sfxSource;
    public AudioClip forestAmbience;
    public AudioClip fireCrackling;
    public AudioClip woodChop;

    void Start()
    {
        bgmSource.clip = forestAmbience;
        bgmSource.loop = true;
        bgmSource.Play();
    }

    public void PlaySFX(AudioClip clip)
    {
        sfxSource.PlayOneShot(clip);
    }
}
```

[Insert Screenshot: AudioManager GameObject showing attached Audio Clips]